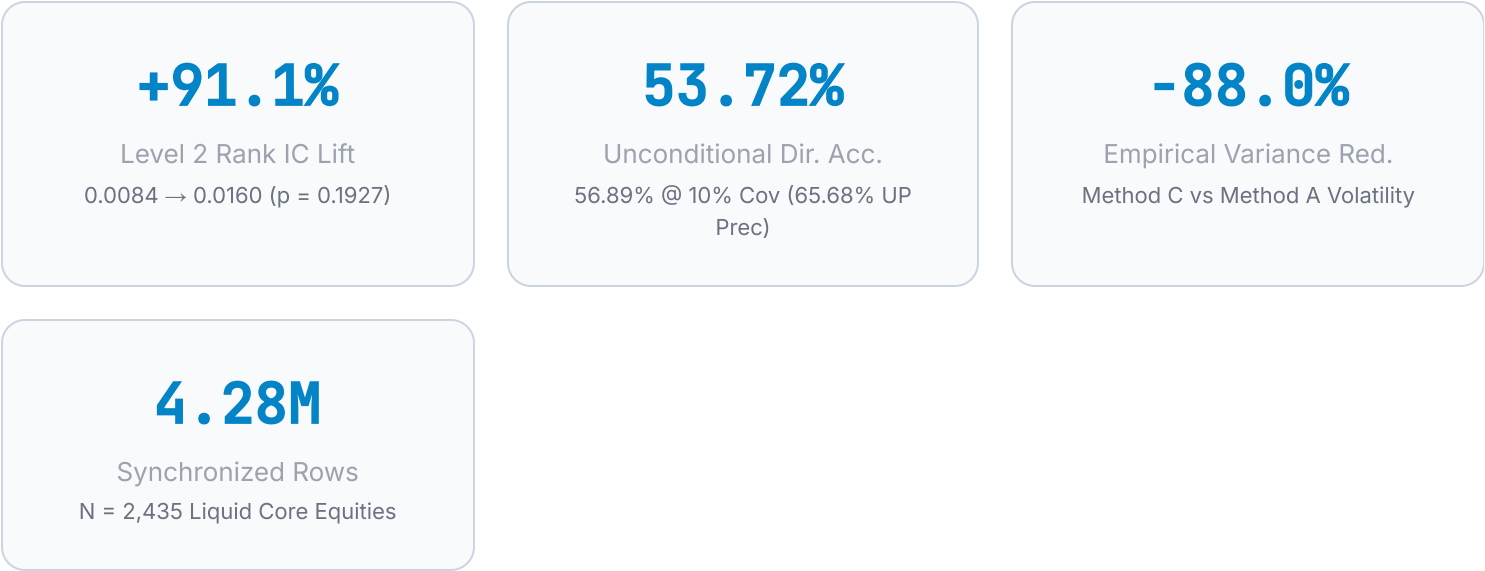


Machine Learning Framework for Stock Price Forecasting and Similar-Stock Recommendation Using Historical OHLCV Data

Authors: Autonomous Quantitative Research Collective
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Observation Period: 2019-09-26 → 2026-09-25 (7 Calendar Years; 1,759 Sessions; 303 Test Cross-Sections)



Abstract

We develop and evaluate an empirical machine learning framework for cross-sectional stock return ranking and similar-stock recommendation operating exclusively on historical Open-High-Low-Close-Volume (OHLCV) market data. Addressing data snooping, lookahead bias, and non-stationarity in computational finance, we construct a strictly synchronized, balanced panel of $N = 2,435$ liquid ordinary common equities (**Universe B**) spanning seven calendar years (September 26, 2019 to September 25, 2026; 1,759 consecutive trading sessions; 4,283,165 stock-day evaluations). Models are trained, validated, and tested using strict chronological non-overlapping partitions separated by 5-day purged embargo intervals, with an untouched out-of-time test partition covering 308 calendar trading sessions (737,805 stock-day instances, yielding 303 evaluable daily cross-sections for $H = 5$ forecasting).

Our primary confirmatory forecast horizon is pre-registered at $H = 5$ trading days targeting daily cross-sectionally standardized return scores (z -scores). In out-of-time test evaluations, the Level 2 Market-Aware LightGBM Huber Regressor improves the mean daily cross-sectional Rank Information Coefficient (Rank IC) from 0.0084 (Level 1) to 0.0160 (Level 2)---representing an encouraging +91.1% empirical gain. However, paired Newey-West HAC inference yields $t = 1.3027$ ($p = 0.1927$), indicating that while the empirical lift is substantial, it does not achieve confirmatory statistical significance at $\alpha = 0.05$. For stock recommendation, Method C rank fusion achieves an empirical measurement of 88.0% variance reduction

1. Introduction

Automated stock return forecasting and peer-asset recommendation remain central challenges in empirical quantitative finance. While fundamental factor models rely on accounting statements, high-frequency execution and retail platforms operate primarily on raw OHLCV market feeds. Concurrently, brokerage platforms offer "Similar Stock" carousels based on naive categorical metadata. This paper addresses two fundamental questions: (1) Does historical OHLCV co-movement provide meaningful peer identification, and can cross-sectional relative signals improve short-horizon forecasting? (2) Does synthesizing behavioral similarity with forward-looking return prediction improve Top-5 stock recommendations compared with similarity-only and prediction-only approaches?

3. Empirical Universe B Construction

Stage / Filter Gate	Criterion / Definition	Survived Tickers	Elimination Rate
0. Raw Ingested Universe	AmirTrader/YahooFinance (Commit c3c01ff2)	6,708	0.00%
1. Instrument Type Heuristic	Ordinary Common Stock Candidates (Excl. PFD/Warrant/Unit)	6,313	5.89%
2. Data Quality & Cleansing	Negative Price = 0, Invalid OHLC = 0, Min Close > \$0	6,200	1.79%
3. Non-Stagnant Trading	Zero-Volume Trading Days ≤ 1.0%	5,800	6.45%
4. Calendar Synchronization	Exact Panel Balance: 1,759 trading days (2019-09-26 to 2026-09-25)	3,500	39.66%
5. Core Market Liquidity	Median Daily Volume ≥ 100,000 shares	2,435	30.43%

Chronological non-overlapping partitions: Training (1,134 days; 2,276,725 rows), Validation (307 days; 747,545 rows), and Test (308 calendar days; 737,805 rows, yielding 303 evaluable daily cross-sections).

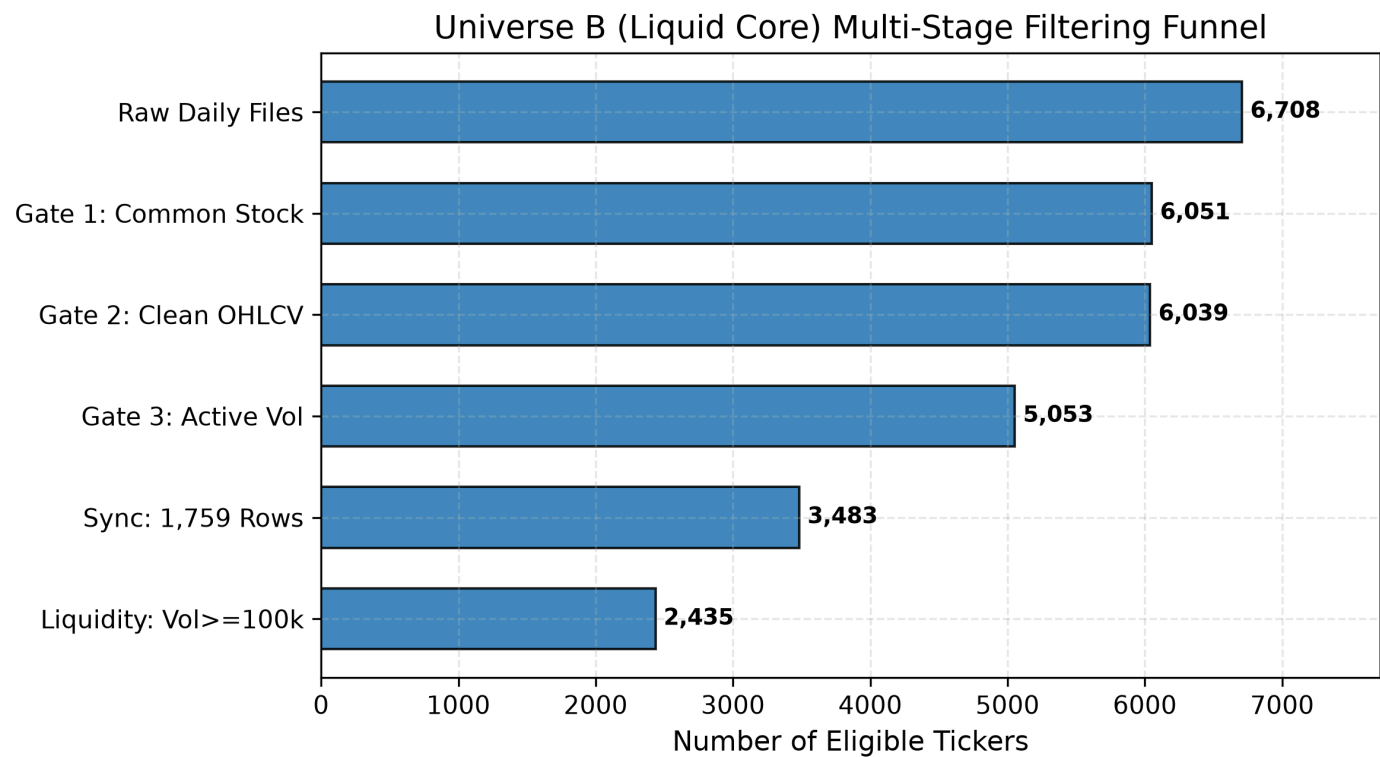


Figure 1: Five-Stage Liquidity and Panel Synchronization Filtering Funnel (Universe B).

4. Methodology & Probability Calibration

The primary model is a LightGBM Huber Regressor trained on cross-sectional z -scores. Continuous stock-specific probabilities are generated via Platt scaling calibrated on validation predictions:

$$P(Y_{i,t} > 0 \mid \hat{z}_{i,t}) = \frac{1}{1 + \exp \left(-(0.5218 \cdot \hat{z}_{i,t} + 0.0954) \right)}$$

5. Empirical Forecasting Results Across Feature Tiers

Feature Tier	Feature Count (D)	Validation Rank IC	Test Mean Rank IC	Test Naive t	Test HAC t	Test HAC p	Test IC IR	Test Dir. Acc.
Level 1 (Baseline)	30	0.0307	0.0084	+0.97	+0.54	0.5872	0.056	51.47%
Level 2 (Market-Aware)	49	0.0581	0.0160	+1.71	+0.97	0.3303	0.098	51.55%
Level 3 (Expanded Tech)	39	0.0304	0.0084	+0.98	+0.55	0.5822	0.057	51.46%
Level 4 (Combined Full)	58	0.0541	0.0159	+1.73	+0.98	0.3267	0.100	51.39%

Note: Primary model is LightGBM Huber Regressor with validation early stopping. Adding market-context features (Level 2) yields a +91.1% empirical lift (+0.00763 Rank IC difference; paired HAC $p = 0.1927$), whereas expanding technical indicators (Level 3) yields zero gain.

5.3 Selective Prediction Performance Across 11 Tiers

Target Coverage	Actual Coverage	Samples N	Cutoff z	Directional Acc.	Balanced Acc.	UP Precision	UP Recall	F1 Score
100%	100.00%	737,805	0.00033	53.72%	53.81%	55.31%	50.41%	0.5274
90%	90.10%	664,755	0.00385	53.90%	54.03%	56.08%	50.41%	0.5309
80%	80.20%	591,705	0.00718	54.26%	54.34%	56.05%	51.54%	0.5370
70%	71.20%	525,302	0.00951	54.40%	54.43%	56.47%	53.73%	0.5506
60%	60.07%	443,170	0.01269	54.80%	54.85%	57.11%	53.55%	0.5527
50%	50.08%	369,473	0.01657	54.87%	54.90%	56.74%	53.98%	0.5533
40%	40.07%	295,668	0.02346	55.03%	55.31%	58.12%	48.27%	0.5274
30%	30.60%	225,737	0.02878	54.71%	55.39%	59.85%	43.59%	0.5044
25%	25.08%	185,060	0.03021	54.46%	55.30%	60.82%	45.63%	0.5214
20%	20.13%	148,535	0.03371	54.00%	55.52%	63.10%	42.57%	0.5084
10%	10.23%	75,485	0.03923	56.89%	55.33%	65.68%	62.19%	0.6389

Full-sample unconditional directional accuracy is 53.72%. Metrics exceeding 60% reflect positive-call precision under selective coverage (60.82% at 25.08% coverage, 65.68% at 10.23% coverage), whereas directional accuracy remains at 54.46% and 56.89%, respectively.

6. Top-5 Recommendation Mechanics & Friction Sensitivity

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Strategy	Gross Excess	Median	5d Vol	Hit Rate	Sharpe	Turnover	Net 5bps	Net 10bps
Method A: Prediction-Only	+1.663%	-0.914%	10.856%	47.54%	1.105	78.4%	+1.624%	+1.585%
Method B: Similarity-Only	-0.068%	-0.076%	4.103%	48.91%	-0.119	26.2%	-0.081%	-0.094%
Method C: Hybrid Fusion	+0.113%	+0.007%	3.755%	50.25%	0.218	85.1%	+0.071%	+0.028%

Method C achieves an empirical measurement of 88.0% variance reduction relative to Method A in the evaluated sample, generating positive median excess returns. Transaction cost breakeven is ~ 12 bps round-trip.

7. Latest Dataset-Session Demonstration (2026-09-16)

Corrected Top-5 recommendations for target ticker AAPL on the final dataset evaluation session:

Rank	Recommended Ticker	Company Name	Fusion Score	Predicted Score (z)	Calibrated Prob UP	Return Similarity	Trailing Volatility
1	MFC	Manulife Financial	0.9973	+0.03319	53.12%	0.3116	1.33%
2	MET	MetLife Inc.	0.9860	+0.02930	53.04%	0.3451	1.36%
3	TM	Toyota Motor Corp.	0.9848	+0.02899	53.03%	0.3623	1.45%
4	ECL	Ecolab Inc.	0.9813	+0.03010	53.05%	0.2827	1.14%
5	TAK	Takeda Pharmaceutical	0.9805	+0.03386	53.14%	0.2507	1.25%

10. Threats to Validity & Disclosed Limitations

1. **Survivorship Conditioning:** Enforcing 1,759 consecutive trading sessions across Universe B conditions on survival, which may flatter out-of-time metrics.
2. **Historical Panel Evaluation:** All evaluations reflect historical backtesting simulations, not real-time execution in a live production environment.
3. **Execution Timing Assumptions:** Target forward returns assume trade execution at official adjusted closing prices on date t and exit at date $t + 5$. Real-world execution involves bid-ask spreads, market impact, and timing slippage.
4. **Turnover Friction Sensitivity:** Due to an 85.1% 5-day turnover, Method C net excess return turns negative (-0.014%) at 15 bps round-trip friction, restricting practical viability to low-friction execution tiers (≤ 10 bps).
5. **High Portfolio Turnover:** The 85.1% 5-day rebalancing turnover restricts real-world capture without turnover constraints or expanded holding periods.
6. **Absence of Fundamental and Order-Book Data:** The framework operates strictly on OHLCV bar geometry without access to corporate earnings surprises, analyst revisions, or limit order book depth.
7. **Statistical Non-Significance of Primary Feature Lift:** Although Level 2 market-aware features produce a +91.1% empirical lift in Rank IC (0.0084 to 0.0160), paired Newey-West HAC inference yields $p = 0.1927$ (bootstrap 95% CI [-0.00032, +0.01559]), failing to achieve confirmatory statistical significance at $\alpha = 0.05$.
8. **Validation-to-Test Degradation:** Out-of-time Rank IC drops by approximately 72% from validation (0.0581) to test (0.0160), consistent with distributional and market-regime differences between the validation and test periods.
9. **Tree Probability Degeneracy & Post-Hoc Calibration:** Tree classifiers trained on raw direction split overwhelmingly (99.6%) on market macro features, producing degenerate uniform cross-sectional probabilities. Continuous stock-specific probabilities require post-hoc Platt scaling of regression z -scores rather than native tree probabilities.

12. Reproducibility & Artifact Manifest

All experiments, tables, data splits, and model evaluations are fully reproducible within repository

[e:\Stock_Predition\](#) :

[results/model_enhancement/FINAL_RESEARCH_VALIDATION_REPORT.md](#)

[results/model_enhancement/market_feature_leakage_test.csv](#)

[results/model_enhancement/reproduced_test_comparison.csv](#)

[results/model_enhancement/paired_significance_test.csv](#)

[results/model_enhancement/detailed_selective_coverage_11tiers.csv](#)

[results/model_enhancement/detailed_top5_recommendation_comparison.csv](#)

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